

Package ‘RSnowflake’

May 22, 2026

Title DBI Interface to 'Snowflake'

Version 0.2.2

Description A DBI-compliant interface to Snowflake (<<https://www.snowflake.com>>).

Connects via Snowflake's SQL API v2 (HTTPS/JSON) as the universal baseline, with no dependency on ODBC, JDBC, or Python. Supports key-pair JWT, programmatic access token (PAT), OAuth, and session token authentication. Optionally uses ADBC (Arrow Database Connectivity) as an accelerator for bulk reads (Arrow-native) and writes (PUT + COPY INTO), activated automatically when adbcsnowflake is installed.

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URL <https://github.com/Snowflake-Labs/RSnowflake>

BugReports <https://github.com/Snowflake-Labs/RSnowflake/issues>

Encoding UTF-8

Roxygen list(markdown = TRUE)

RoxygenNote 7.3.2

Depends R (>= 4.1.0)

Imports DBI (>= 1.2.0),
methods,
httr2 (>= 1.0.0),
jsonlite,
rlang (>= 1.0.0),
cli

Suggests openssl,
jose,
RcppTOML,
RcppSimdJson,
nanoarrow (>= 0.3.0),
adbcdrivermanager,
adbcsnowflake,
dbplyr (>= 2.3.0),
dplyr,
duckdb,
blob (>= 1.2.0),
hms (>= 1.0.0),
bit64,
reticulate,

snowflakeauth,
 DBItest (>= 1.7.0),
 testthat (>= 3.0.0),
 mockr,
 withr,
 knitr,
 rmarkdown

VignetteBuilder knitr

Config/testthat/edition 3

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sf_connect	<i>Connect to Snowflake using a profile name</i>
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Description

Convenience wrapper around `dbConnect(Snowflake(), name = ...)`.

Usage

```
sf_connect(name = NULL, ...)
```

Arguments

name	Profile name from <code>connections.toml</code> .
...	Additional arguments passed to <code>dbConnect</code> .

Value

A `SnowflakeConnection`.

Snowflake*Create a SnowflakeDriver instance*

Description

Create a SnowflakeDriver instance

Usage

```
Snowflake()
```

Value

A SnowflakeDriver singleton.

Examples

```
## Not run:
drv <- Snowflake()
con <- dbConnect(drv, account = "myaccount", name = "default")

## End(Not run)
```

SnowflakeConnection-class*SnowflakeConnection*

Description

An S4 class representing a connection to Snowflake via the SQL API v2. Created by `dbConnect()` with a `Snowflake()` driver.

Usage

```
## S4 method for signature 'SnowflakeConnection'
show(object)

## S4 method for signature 'SnowflakeConnection'
dbIsValid(dbObj, ...)

## S4 method for signature 'SnowflakeConnection'
dbDisconnect(conn, ...)

## S4 method for signature 'SnowflakeConnection'
dbGetInfo(dbObj, ...)

## S4 method for signature 'SnowflakeConnection,character'
dbSendQuery(conn, statement, params = NULL, ...)

## S4 method for signature 'SnowflakeConnection,character'
```

```
dbSendStatement(conn, statement, params = NULL, ...)

## S4 method for signature 'SnowflakeConnection,character'
dbGetQuery(conn, statement, params = NULL, ...)

## S4 method for signature 'SnowflakeConnection,character'
dbExecute(conn, statement, params = NULL, ...)

## S4 method for signature 'SnowflakeConnection,character'
dbGetQueryArrow(conn, statement, params = NULL, ...)

## S4 method for signature 'SnowflakeConnection,character'
dbSendQueryArrow(conn, statement, params = NULL, ...)

## S4 method for signature 'SnowflakeConnection'
dbListTables(conn, ...)

## S4 method for signature 'SnowflakeConnection,character'
dbExistsTable(conn, name, ...)

## S4 method for signature 'SnowflakeConnection,character'
dbListFields(conn, name, ...)

## S4 method for signature 'SnowflakeConnection'
dbListObjects(conn, prefix = NULL, ...)

## S4 method for signature 'SnowflakeConnection,character'
dbReadTable(conn, name, ...)

## S4 method for signature 'SnowflakeConnection,character,data.frame'
dbWriteTable(
  conn,
  name,
  value,
  overwrite = FALSE,
  append = FALSE,
  row.names = FALSE,
  ...
)

## S4 method for signature 'SnowflakeConnection,character'
dbCreateTable(conn, name, fields, ..., row.names = NULL, temporary = FALSE)

## S4 method for signature 'SnowflakeConnection,character'
dbAppendTable(conn, name, value, ..., row.names = NULL)

## S4 method for signature 'SnowflakeConnection,character'
dbRemoveTable(conn, name, ...)

## S4 method for signature 'SnowflakeConnection,character'
dbQuoteIdentifier(conn, x, ...)
```

```

## S4 method for signature 'SnowflakeConnection,character'
dbQuoteString(conn, x, ...)

## S4 method for signature 'SnowflakeConnection,SQL'
dbQuoteString(conn, x, ...)

## S4 method for signature 'SnowflakeConnection,SQL'
dbQuoteIdentifier(conn, x, ...)

## S4 method for signature 'SnowflakeConnection,SQL'
dbUnquoteIdentifier(conn, x, ...)

## S4 method for signature 'SnowflakeConnection'
dbQuoteLiteral(conn, x, ...)

## S4 method for signature 'SnowflakeConnection'
dbDataType(dbObj, obj, ...)

## S4 method for signature 'SnowflakeConnection'
dbBegin(conn, ...)

## S4 method for signature 'SnowflakeConnection'
dbCommit(conn, ...)

## S4 method for signature 'SnowflakeConnection'
dbRollback(conn, ...)

## S4 method for signature 'SnowflakeConnection'
dbWithTransaction(conn, code, ...)

```

Arguments

object	A SnowflakeConnection object (for show).
dbObj	A SnowflakeConnection object.
...	Additional arguments (currently ignored).
conn	A SnowflakeConnection object.
statement	SQL statement string.
params	Optional parameter list for parameterized queries.
name	Table name (character).
prefix	An Id object indicating the hierarchy level to list (NULL for databases, 1-component for schemas, 2-component for tables).
value	A data.frame.
overwrite	Logical. Drop existing table first?
append	Logical. Append to existing table?
row.names	Ignored (always FALSE for Snowflake).
fields	A data.frame or named character vector of column types.
temporary	Logical. Create a temporary table?
x	Character to quote.
obj	An R object to map to a Snowflake SQL type.
code	Code to execute within the transaction.

Slots

account Account identifier.
 user Username.
 database Default database.
 schema Default schema.
 warehouse Default warehouse.
 role Default role.
 .auth Auth list (type, token, token_type, plus params for refresh).
 .state Environment holding mutable state (valid, in_transaction, session_info).

SnowflakeDriver-class *SnowflakeDriver*

Description

An S4 class representing the Snowflake DBI driver.

Usage

```

## S4 method for signature 'SnowflakeDriver'
dbGetInfo(dbObj, ...)

## S4 method for signature 'SnowflakeDriver'
dbIsValid(dbObj, ...)

## S4 method for signature 'SnowflakeDriver'
dbUnloadDriver(drv, ...)

## S4 method for signature 'SnowflakeDriver'
show(object)

## S4 method for signature 'SnowflakeDriver'
dbDataType(dbObj, obj, ...)

## S4 method for signature 'SnowflakeDriver'
dbConnect(
  drv,
  account = NULL,
  user = NULL,
  token = NULL,
  private_key_path = NULL,
  authenticator = NULL,
  database = "",
  schema = "",
  warehouse = "",
  role = "",
  name = NULL,
  ...

```

```
)

## S4 method for signature 'SnowflakeDriver'
dbCanConnect(drv, ...)
```

Arguments

dbObj	A SnowflakeDriver object.
...	Additional arguments (ignored).
drv	A SnowflakeDriver , or missing (uses default).
object	A SnowflakeDriver object (for show).
obj	An R object to map to a Snowflake SQL type.
account	Snowflake account identifier (e.g. "myaccount").
user	Snowflake username.
token	Explicit bearer token (PAT or session token).
private_key_path	Path to PEM private key for JWT auth.
authenticator	Auth method ("SNOWFLAKE_JWT", "OAUTH", etc.).
database	Default database.
schema	Default schema.
warehouse	Default warehouse.
role	Default role.
name	Profile name from connections.toml.

SnowflakeResult-class *SnowflakeResult*

Description

An S4 class representing the result of a query against Snowflake. Created by [dbSendQuery\(\)](#) or [dbSendStatement\(\)](#).

Usage

```
## S4 method for signature 'SnowflakeResult'
show(object)

## S4 method for signature 'SnowflakeResult'
dbIsValid(dbObj, ...)

## S4 method for signature 'SnowflakeResult'
dbClearResult(res, ...)

## S4 method for signature 'SnowflakeResult'
dbFetch(res, n = -1, ...)

## S4 method for signature 'SnowflakeResult'
```

```

dbHasCompleted(res, ...)

## S4 method for signature 'SnowflakeResult'
dbFetchArrow(res, ...)

## S4 method for signature 'SnowflakeResult'
dbFetchArrowChunk(res, ...)

## S4 method for signature 'SnowflakeResult'
dbGetRowCount(res, ...)

## S4 method for signature 'SnowflakeResult'
dbGetRowsAffected(res, ...)

## S4 method for signature 'SnowflakeResult'
dbGetStatement(res, ...)

## S4 method for signature 'SnowflakeResult'
dbColumnInfo(res, ...)

## S4 method for signature 'SnowflakeResult'
dbGetInfo(dbObj, ...)

## S4 method for signature 'SnowflakeResult'
dbBind(res, params, ...)

```

Arguments

object	A SnowflakeResult object (for show).
dbObj	A SnowflakeResult object.
...	Additional arguments (currently ignored).
res	A SnowflakeResult object.
n	Number of rows to fetch. -1 (default) fetches all.
params	A list of parameter values for parameterized queries.

Slots

connection	The parent SnowflakeConnection .
statement	The SQL statement string.
.resp_body	The raw JSON response from the first partition.
.meta	Parsed metadata (columns, partitions, handle).
.state	Environment holding mutable state (fetched, partition, valid, etc.).

sqlData*Convert a data.frame to SQL-ready strings*

Description

Convert a data.frame to SQL-ready strings

Usage

```
sqlData(con, value, row.names = FALSE, ...)
```

Arguments

con	A SnowflakeConnection object.
value	A data.frame.
row.names	Ignored.
...	Additional arguments (ignored).

Value

A data.frame of character columns.

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